Prepared for Mr. Gotham Ngosa



KAPASA MAKASA UNIVERSITY

CYBER SECURITY

CYBERCRIME, INVESTIGATION & FORENSIC II (CYS340)

EXPECTED QUESTIONS

Questions by James Soko

UNIT 1 – OVERVIEW

1. Define the term cybercrime. [2 Marks]
2. According to Act no. 2 of 2021 PART I (Preliminary Provision) "Law Enforcement" means – [5 Marks]
3. List three job duties of Cybercrime Investigators. [3 Marks]
4. What is Cybercrime Investigation? [2 Marks]
5. List and explain the top 5 cybercrimes. [5 Marks]
6. Mention three ways on how you can protect against cybercrimes. [3 Marks]
7. List and explain the 5 types of cyber criminals you know. [5 Marks]

TOTAL: [25 Marks]

8. What is Forensics? [2 Marks]
9. What do you understand by the term Digital Forensics? [2 Marks]
10. List and explain the 6 cybercrime investigation techniques. [6 Marks]
11. Cybercrime investigation require the use of automated tools to carry out the task in different fields. Complete the table below by listing the tools used. [15 Marks]

DIGITAL FORENSIC SOFTWARE [2]	NETWORK ANALYSIS TOOLS [3]	MALWARE ANALYSIS TOOLS [3]	PASSWORD RECOVERY TOOLS [4]	SOCIAL MEDIA ANALYSIS TOOLS [3]
FTK				

TOTAL: [25 Marks]

UNIT 2 – DIGITAL FORENSICS

1. What is forensics? [1 Mark]
2. What is computer forensics? [2 Marks]
3. List and explain the characteristics of computer forensics. [5 Marks]
4. Explain at least 3 needs of computer forensics. [3 Marks]
5. Explain at least 3 digital investigation rules and regulations. [3 Marks]
6. List and explain the 6 roles and use of digital forensics by giving an example (s) each. [6 Marks]

Total: [20 Marks]

7. Explain the key steps in a forensic investigation. [10 Marks]
8. Give at least 3 tasks that are performed by forensic investigators. [3 Marks]
9. What does the Locard's exchange principle states? [2 Marks]
10. List and explain at least 3 organizations involved in digital forensics. [3 Marks]
11. List and explain at least 4 computer forensic tools. [4 Marks]
12. What is a digital footprint? [1 Mark]
13. List and explain the two types of digital footprints by giving examples each. [2 Marks]

Total: [25 Marks]

UNIT 3 – FILE CARVING AND DATA RECOVERY

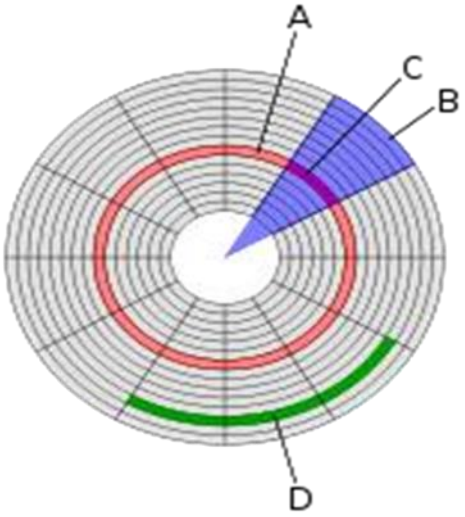
1. What is computer data? [1 Mark]
2. What is data recovery? [2 Marks]
3. Mention at least 3 tools used in data recovery. [3 Marks]
4. Explain the 2 bad tracks of a hard disk. [2 Marks]
5. List and explain at least 3 data recovery methods. [3 Marks]
6. Give 6 reasons for data loss. [6 Marks]
7. What is a hard disk? [1 Mark]
8. Explain the data storage region of a hard drive using a table. [5 Marks]
9. Explain the 2 types of file deletion. [2 Marks]

Total: [25 Marks]

10. Explain the 2 types of high-level formatting. [2 Marks]
11. Give 4 difference of FAT and NTFS in a table format. [4 Marks]
12. Give two major deletion techniques. [2 Marks]

UNIT 4 – HARD DISK DRIVE

13. List and explain the 3 major components of a hard drive. [3 Marks]
14. Complete the disk structure below. [4 Marks]



15. List and explain the 4 partitions of an HDD. [4 Marks]
16. List and explain the 3 types of interfacing in an HDD. [3 Marks]
17. Tabulate the HDD capacity and its size information in s table format. [3 Marks]

Total: [25 Marks]

UNIT 5 – MULTI MEDIA FORENSICS

1. List at least 5 classification/elements of multimedia. [5 Marks]
2. What is multimedia? [2 Marks]
3. Mention the 4 major use area of multimedia. [4 Marks]
4. Differentiate between mono and stereo sound by giving examples. [2 Marks]
5. How different is multimedia forensics from other forms of forensics? [2 Marks]
6. Mention at least 2 ways in how one can authenticate the video or audio recording. [2 Marks]
7. List and explain two types of video and audio forms. [2 Marks]
8. Video and audio recorders came in two general types namely…. [2 Marks]
9. What important evidence should be recorded when a forensic investigator is using a camera? [4 Marks]

TOTAL: [25 Marks]

10. Who conducts the video and audio analysis and how did they categorize them? [5 Marks]
11. How is the video and audio analysis performed? [3 Marks]
12. List and explain at least 4 video enhancement techniques. [4 Marks]
13. Mention and explain the 2 major audio enhancement techniques. [4 Marks]
14. List at least any 2 limitations associated with video and audio in multimedia forensics. [2 Marks]

TOTAL: [20 Marks]

UNIT 6 – MALWARE

1. Define the term Malware [2 Marks]
2. What does Mal mean in Spanish? [1 Mark]
3. In which year was the term malware coined? [1 Mark]
4. In which year did malware became frequently used? [1 Mark]
5. What was the name of the first malware created? [1 Mark]
6. When was the malware mentioned in Q5 created? [1 Marks]
7. What was the first Anti-Virus software created to detect and destroy the malware mentioned in Q5? [1 Mark]
8. List at least three (3) purposes why people create/distribute malware. [3 Marks]
9. How can one distribute malware? [4 Marks]
10. Using a visual diagram, explain the five (5) stages of malware attack. [5 Marks]

TOTAL: [20 Marks]

11. List the three (3) major types of malwares. [3 Marks]
12. Define the types of malwares mentioned in Q11. [6 Marks]
13. List and explain at least three (3) types of Viruses. [6 Marks]
14. List and explain the two (2) types of computer worms. [4 Marks]

TOTAL: [19 Marks]

15. List and explain at least four (4) types of Trojan Horses. [8 Marks]
16. Explain how an attacker launches a Ransomware attack. [4 Marks]
17. List and explain the three (3) types of Ransomwares. [6 Marks]
18. Where was Ransomware discovered? [1 Mark]
19. In which year (s) was the Ransomware attack discovered? [1 Mark]
20. In which year was the Crypto-ransomware discovered? [1 Mark]
21. Using Anti-Malware/Anti-Virus software's is one way to protect your computer from viruses. Windows does offer such security in real time.
    a) What do was the security feature used by Windows XP, Vista and 7? [1 Mark]
    b) What is the security feature currently used by Windows 8, 10 and 11? [1 Mark]
    c) Which Windows software is used to remove malicious software from the system? [1 Mark]
    d) List at least three (3) commercial Anti-Virus software's. [1 Mark]

TOTAL: [25 Marks]

22. Anti-viruses come in two types, write few notes and highlight how the differ from one another. [4 Marks]

UNIT 7 – CLOUD FORENSICS

1. Define the term cloud forensics. [2 Marks]
2. List and explain the 4 types of cloud deployment models. [4 Marks]
3. Cloud offers five (5) services and these include Container as a Service (CaaS) and Functions as a Service (FaaS). Using a table format, differentiate the other three (3) services. [6 Marks]
4. List the five (5) essential characteristics of cloud computing. [5 Marks]
5. What is cloud forensics? [2 Marks]

TOTAL: [23 Marks]

6. List and explain the five (5) actors involved in cloud computing. [5 Marks]
7. Hight the three (3) scenarios that might exit among the cloud actors. [3 Marks]
8. What are Service Level Agreements (SLAs)? [2 Marks]
9. List and explain the three (3) categories provided by an actor who negotiates relationships on behalf of the client using the cloud. [3 Marks]
10. Who is a transport agent? [2 Marks]

UNIT 8 – MEMORY FORENSICS

1. Write few notes about the RAM and why is it necessary in memory forensics? [2 Marks]
2. What is memory forensics? [2 Marks]
3. Hight the three (3) major deference's between memory forensics and hard disk forensics. [3 Marks]
4. Give four reasons why memory forensics is needed in certain situations that hard disk forensics. [4 Marks]

TOTAL: [25 Marks]

5. List and explain the two (2) types of data acquisition methods in memory forensics by stating their advantages and disadvantages. [4 Marks]
6. List the five (5) types of formats acquired in memory forensics. [5 Marks]
7. List and write few notes on any three (3) tools used in memory forensics. [3 Marks]

TOTAL: [12 Marks]

Prepared for Mr. Gotham Ngosa

I have left out some questions on Cloud Computing, so make sure to study the handout given by our lecture. All the best!!!